HANOI UNIVERSITY OF
SCIENCE AND TECHNOLOGY

SCHOOL OF INFORMATION AND COMMUNICATION TECHNOLOGY

INTERNSHIP TECHNICAL REPORT

BudgetRAG / MemAlign-Qwen

Resource-Aware Retrieval and Context Allocation for Grounded LLM Inference

Student: Kieu Son Tung (20235571)

Class: DSAI 02 - K68

Supervisor: Assoc. Prof. Le Thanh Huong

Department: Information Systems

Repository: https://github.com/sontungkieu/true-chat

June 5, 2026

Abstract

Long-context Retrieval-Augmented Generation (RAG) improves grounding by giving the generator more retrieved evidence, but it also increases prompt length, latency, and estimated decoder KV-cache memory. This report studies resource-aware retrieval and context allocation: for a query and a set of retrieved candidates, the system must select a retrieval-context action that preserves grounded answer quality while reducing token, latency, and estimated KV-cache costs. The implementation begins with a benchmark foundation for retriever registration, benchmark execution, and retrieval metrics. It then adds BudgetRAG context policies, deterministic adaptive profiles, generation feedback, RAGAS/MiMo answer labels, RLAIF-style answer and context labels, reward/preference construction, and offline selector baselines. The main empirical evidence is diagnostic: 192 MiMo answer labels, 192 merged MiMo context labels with 177 clean usable rows, context sufficiency rate 0.591, mean selected chunks 1.276 versus mean irrelevant chunks 3.708, and context-reward ablations that change 140 of 192 reward rows. Retriever-diversity logging now includes a 2250-row retrieval-only coverage run and a 1500-row standard MiMo V2.5 A1-medium generation gate over BM25, graph-BM25, and hybrid-RRF. A1 has 1500/1500 valid answer labels, 1460 clean usable answer labels, and a stratified 600-row context-label pass with 548 clean usable labels, context sufficiency rate 0.6985, mean context quality 0.6431, and mean evidence support 0.6567. Graph-BM25 is strongest at answer level, while the stratified context labels favor hybrid-RRF on sufficiency/support and graph-BM25 on efficiency/noise. The report therefore treats retriever ranking as provisional and query-conditioned. It does not claim human preference learning, online reinforcement learning, DPO/PPO/GRPO training, or deployed KV pruning.

Contents

1 Introduction

Retrieval-Augmented Generation (RAG) combines external retrieval with language-model generation so that answers can be grounded in retrieved evidence rather than only in parametric memory [8]. The practical baseline is often simple: retrieve the top-k documents and place most of their content into the prompt. That baseline is easy to implement, but it is not resource-aware. Retrieved context can contain irrelevant chunks, longer prompts increase latency and cost, and long sequence lengths increase decoder KV-cache memory for local LLM inference.

BudgetRAG / MemAlign-Qwen addresses this trade-off. The core question is: for a given query and a set of retrieved candidates, which retrieval strategy, context policy, and context budget should be selected to maximize grounded answer quality while controlling token count, latency, and estimated KV-cache footprint?

The MemAlign-Qwen motivation is local and privacy-constrained inference. In later phases, private documents should be processed by trusted local models rather than remote APIs. In the current phase, BudgetRAG reduces context before inference and uses RLAIF-style context labels as future evidence for local Qwen profiling and possible KV evidence masks. The current report is careful about the boundary: the system is an offline research pipeline, not a deployed production policy.

1.1 Contributions

This internship produced six concrete contributions:

1. A reproducible RAG/BudgetRAG benchmark pipeline with fixed context policies, adaptive profiles, traceable action rows, and curated reports.
2. An RLAIF-style feedback layer that records answer-level labels, context-level sufficiency labels, and direct pairwise judge audits.
3. A reward/preference construction pipeline with explicit quality, evidence-support, token, latency, KV, unsupported-claim, and error components.
4. Held-out offline selector baselines that compare cheapest, average, smoothed, learned linear, and oracle-logged policies without replacing the runtime heuristic.
5. Full MiMo context-label evidence showing that retrieved contexts often contain only a small number of useful chunks and several irrelevant chunks.
6. Analytical Qwen KV-cache estimates and a deployment-oriented vLLM model-benchmark workflow to connect BudgetRAG with future local inference experiments.

2 Problem Formulation

For a query q and a candidate set C returned by one or more retrievers, the system chooses an action:

$$a = (\text{retrieval strategy}, \text{fusion strategy}, \text{context policy}, \text{budget}, \text{adaptive profile}, \text{generator model}).$$

The objective is an offline reward that trades off answer quality and evidence support against resource costs:

$$R = w_q Q + w_s S - w_t T - w_l L - w_k K - P_{\text{unsupported}} - P_{\text{error}}.$$

Here Q is answer quality/correctness, S is evidence or context support, T is normalized token cost, L is normalized latency, K is normalized estimated KV-cache cost, and the penalty terms capture unsupported claims and failed/invalid generations or labels.

The state includes query features, retrieval diagnostics, action metadata, and cost estimates. The evaluation is **offline logged-candidate evaluation**: a selector can only choose among actions that were already logged for a query. It is not online RL, and selector artifacts explicitly keep `runtime_default_replacement=false`.

3 Related Work

3.1 RAG and RAG Evaluation

RAG introduced a practical way to combine retrieval with neural generation for knowledge-intensive tasks [8]. BEIR showed that retrieval quality varies across heterogeneous domains and that no single retriever should be treated as universally best [14]. RAGAS adds automated RAG evaluation signals such as answer relevance and faithfulness [5]. In this project, BEIR-style metrics motivate retrieval benchmarking, while RAGAS is used as an initial proxy before richer MiMo answer labels are introduced.

3.2 Prompt and Context Compression

LLMLingua, LongLLMLingua, and Selective Context study prompt/context compression for efficient LLM inference [6, 7, 10]. BudgetRAG differs in formulation: context compression is not a standalone preprocessing step, but part of a retrieval-context action evaluated under answer quality, context sufficiency, token cost, latency, and estimated KV cost.

3.3 KV-Cache Compression and Pruning

H2O, StreamingLLM, KIVI, SnapKV, and PyramidKV reduce runtime memory or attention cost through cache selection, quantization, or pruning [16, 15, 12, 11, 2]. BudgetRAG currently operates before inference by selecting a smaller or cleaner context. Runtime KV pruning is future work; context-level RLAIF labels may later provide evidence-token masks.

3.4 Graph Retrieval, RLAIF, and Contextual Bandits

GraphRAG highlights graph-based retrieval for local-to-global query-focused summarization [4]. Hybrid retrieval and Reciprocal Rank Fusion (RRF) are relevant for combining retrieval strategies [3]. RLAIF and preference learning use AI feedback to construct reward or preference signals [1]; DPO is a later preference-optimization method, not implemented here [13]. Because the BudgetRAG action is a one-step query-time decision, offline contextual bandit framing is more appropriate than PPO/GRPO in the current data-limited phase [9].

3.5 Positioning of This Work

Research line	What it solves	Remaining gap	BudgetRAG position
RAG evaluation	Measures retrieval and answer quality.	Does not choose a resource-aware context action.	Uses retrieval and answer metrics as logged feedback.
Prompt compression	Shortens prompts before generation.	Often treats compression separately from retrieval policy.	Makes compression a retrieval-context action.
KV-cache methods	Reduce runtime cache memory or attention cost.	Usually operate inside model inference.	Works before inference; KV pruning remains future work.
Graph/hybrid retrieval	Improves evidence discovery and fusion.	Needs action-level allocation evidence.	Treats retriever choice as part of future action coverage.
RLAIF/preference learning	Builds AI-feedback rewards/preferences.	General alignment methods do not target RAG context budgets.	Uses AI feedback for answer and context supervision.
Contextual bandits	Learn one-step action selection.	Require enough logged action coverage.	Evaluates offline selectors over logged candidates only.

4 System Overview

The project pipeline is:

`Query` → retriever / graph / hybrid retrieval → BudgetRAG context policy → generator → answer/context feedback → reward/preference rows → offline selector evaluation

The main implementation components are:

- `rag-bench`: benchmark, BudgetRAG, and RLAIF CLI.
- Retriever registry: BM25 plus scaffolded graph/hybrid strategies.
- Context policies: `legacy`, `char-budget`, `per-doc-budget`, `score-density`, `sentence-trim`, `evidence-aware`, and `adaptive-heuristic`.
- Adaptive profiles: conservative, balanced, and aggressive.
- RLAIF artifacts: `rlaif_actions.jsonl`, `rlaif_feedback.jsonl`, `rlaif_rewards.jsonl`, and `rlaif_preferences.jsonl`.
- Labelers: `rlaif-label-answers`, `rlaif-label-contexts`, and `rlaif-label-pairs`.
- Selectors: fixed, cheapest, best average, family-smoothed, shrinkage-smoothed, linear reward model, smoothed linear selector, and oracle logged.

4.1 Method Summary

The method can be read as three connected stages.

BudgetRAG context allocation. A query first produces candidate evidence through a registered retriever. A context action then decides how much evidence to keep and how to trim it. The action may preserve the legacy full-context behavior, impose character budgets, distribute budget per document, prefer high score-density chunks, trim at sentence boundaries, or use deterministic adaptive profiles.

RLAIF feedback and reward construction. After generation, answer-level and context-level judges label the logged row. Answer labels estimate correctness, support, unsupported claims, and refusal behavior. Context labels estimate whether the retrieved context is sufficient, which chunks are useful, and which chunks are redundant or irrelevant. These labels are converted into scalar rewards and pairwise preferences with guardrails so missing or ambiguous feedback is not treated as zero quality.

Offline selector evaluation. Selector policies are trained or summarized only on the logged train split and are evaluated on held-out query groups. The selector can choose only among candidate actions already logged for that query, which makes the experiment an offline logged-candidate study rather than an online RL deployment.

The design keeps raw experiment outputs under ignored `benchmark_results/`; committed reports are curated summaries. This prevents raw judge outputs, downloaded Kaggle working directories, or secrets from becoming part of the repository history.

5 Experimental Setup

The experiments are organized into retrieval-only, generation-feedback, and RLAIF-feedback stages. This separation keeps the evidence for each claim clear: the early phases validate benchmark plumbing and context budgeting, while the later phases evaluate answer and context quality.

5.1 Datasets and Logged Rows

The completed empirical results use SciFact-style BudgetRAG logs. Phase 1B and Phase 1C retrieval-only runs use BM25 on SciFact with generation skipped. Phase 1C.3 and Phase 1D use 192 joined action-query rows derived from BudgetRAG generation outputs and RAGAS post-hoc rows.

Item	Count
Phase 1D joined action-query rows	192
RAGAS source action files	64
Query groups in selector smoke	55
Action signatures	77
Full MiMo answer labels	192
Full MiMo context labels	192
Clean usable context labels	177
Targeted multi-judge audit rows	60
A1 retriever-diverse action rows	1500
A1 clean answer labels	1460
A1 stratified context labels	600
A1 clean context labels	548

The original Phase 1D action log is too BM25-heavy to support strong retrieval-strategy allocation claims by itself. The A1 retriever-diverse extension adds BM25, graph-BM25, and hybrid-RRF as separate logged action families, but the final retriever ranking remains provisional because context labels cover a stratified 600/1500 subset.

A small sampled HotpotQA Kaggle evaluation is included later as supplementary multi-hop stress evidence. It is not part of the main Phase 1D selector training/evaluation set, and it is not reported as a full HotpotQA benchmark claim.

5.2 Models and Judges

The generation/action matrix includes MiMo, Llama 3.1 8B, and Qwen3 32B rows. RAGAS answer relevancy provides the initial proxy label. MiMo provides full answer-level and context-level labels. DeepSeek v4 Flash and Groq Qwen3 32B provide secondary labels in a targeted context audit.

The judge prompts are constrained to logged question, answer, and context. They are not allowed to browse or use external knowledge. Invalid JSON, missing answers, missing context, and ambiguous judge outputs are recorded as ambiguous/null-score rows rather than converted into zero-quality labels.

5.3 Split Rule and Selector Evaluation

Selector evaluation uses deterministic held-out splits at query level:

$$\text{split key} = \texttt{benchmark + query\_id}.$$

This prevents action-row leakage: all actions for the same query remain in the same split. Preferences that cross train/eval boundaries are dropped and counted in the split manifest. Selector costs are logged/offline normalized costs; a deployable pre-generation selector would require estimated token/KV costs and predicted latency features.

5.4 Reward Weights

The main scalar reward uses quality weight 0.75, evidence-support weight 0.10, token weight 0.05, latency weight 0.05, and KV weight 0.05, with explicit penalties for unsupported claims and errors. Context-label reward is opt-in and non-default. The reported context penalty weights are 0.25, 0.50, and 1.00.

6 Phase 1A: Benchmark Foundation

Phase 1A created the benchmark foundation: a retriever registry, benchmark CLI, retrieval metrics, and traceable output records. The metric layer includes hit@k, MRR, nDCG, latency, and context size/cost fields. This phase made later BudgetRAG experiments reproducible: fixed policies, adaptive profiles, and RLAIF builders all consume the same output format.

Phase 1A established the reproducible benchmarking infrastructure. Quantitative reporting starts from Phase 1B/1C, where committed reports contain concrete matrix outputs.

Phase	Goal	What was implemented	Main finding
1A	Reproducible RAG benchmarking	Retriever registry, benchmark CLI, retrieval metrics, traceable records.	Later phases could reuse one output schema.
1B	Fixed context budgeting	Legacy, character, per-document, score-density, sentence-trim, and evidence-aware policies.	Compression/KV savings could be measured separately from answer quality.
1C	Deterministic adaptation	Adaptive heuristic plus conservative, balanced, and aggressive profiles.	Conservative settings were stable but too cautious; profiles increased action diversity.
1C.3	Generation feedback	Multi-model generation rows, MiMo long-context runs, HotpotQA sampled Kaggle path.	Answer labels were needed; retrieval-only metrics were insufficient.
1D	RLAIF selector	Answer/context/pairwise labels, rewards/preferences, held-out selector baselines.	Context-level labels materially changed rewards and exposed evidence insufficiency.

7 Phase 1B: BudgetRAG Fixed Context Policies

Phase 1B introduced fixed context policies. `legacy` preserves the older behavior, `char-budget` clips total characters, `per-doc-budget` avoids one long document dominating the prompt, `score-density` favors high score per length, `sentence-trim` avoids cutting inside sentences, and `evidence-aware` keeps lexical/query-aware evidence spans. Importantly, `evidence-aware` is not an answer-aware entailment verifier.

The Phase 1B smoke run used 10 SciFact queries, BM25, top-k 3, skipped generation, and a Qwen2.5-14B KV profile. The 1000-character variants kept about 1000 characters with compression around 0.216 and analytical KV savings around 900 units in the report. Quality columns are blank because generation was skipped.

8 Phase 1C: Adaptive Profiles

Phase 1C replaced a purely fixed context budget with a deterministic adaptive heuristic. The selector uses retrieval diagnostics such as score gap, normalized entropy, confidence, and long-document dominance.

Phase 1C.1 expanded validation to 50 SciFact BM25 queries. The heuristic selected `evidence-aware` for 48 queries and `per-doc-budget` for 2, but still selected budget 4000 for all 50 queries. This showed stable metadata but an overly conservative policy.

Phase 1C.2 added conservative, balanced, and aggressive profiles. Balanced and aggressive profiles created more diverse budget choices. For example, with matrix budget 1000, conservative kept 4000 characters for all 50 queries, balanced averaged 2260 kept characters, and aggressive averaged 999.8 kept characters. This was still a deterministic heuristic baseline, not a learned selector.

9 Phase 1C.3: Generation Feedback Layer

Retrieval-only metrics cannot determine whether compressed context still supports a grounded answer. Phase 1C.3 added a generation feedback layer. RAGAS answer relevancy was first joined back to BudgetRAG `query_results.jsonl`; then MiMo answer labels were added for richer answer correctness, evidence support, faithfulness, and unsupported-claim penalties.

The Phase 1D selector-smoke input contained 192 joined action-query rows from 64 action files. Model coverage was MiMo 96 rows, Llama 3.1 8B 48 rows, and Qwen3 32B 48 rows. Context-policy coverage was adaptive-heuristic 96, evidence-aware 48, and legacy 48. Budget coverage included 4000:48, 8000:48, 1000:36, 2000:36, 16000:12, and 32000:12.

The full MiMo answer-label run produced 192 labels, 192 valid JSON rows, 0 invalid JSON rows, 25 ambiguous labels, 1 judge error, and 191 scored labels. The Pearson correlation between MiMo quality and RAGAS answer relevancy was 0.277. This low correlation supports the interpretation that RAGAS answer relevancy and MiMo answer quality measure related but not identical signals.

10 Phase 1D: RLAIF Reward and Preference Pipeline

Phase 1D converts BudgetRAG logs into normalized RLAIF-style datasets:

Command	Role
`rlaif-build`	Convert BudgetRAG rows into normalized actions and feedback.
`rlaif-label-answers`	Label answer correctness, support, faithfulness, and unsupported claims.
`rlaif-label-contexts`	Label context sufficiency, useful chunks, redundant chunks, irrelevant chunks, and missing evidence.
`rlaif-reward`	Build scalar rewards and pairwise preferences with quality guardrails.
`rlaif-split`	Split by `benchmark + query_id` to avoid row-level leakage.
`rlaif-train/eval`	Train and evaluate offline selector baselines.
`rlaif-label-pairs`	Directly judge action pairs for reward calibration audits.

The reward builder preserves provenance and missing reasons. Missing, ambiguous, or invalid labels do not become score zero. Preferences are generated within comparable query groups, with guardrails that reject efficiency-only wins when answer quality drops too much.

RLAIF in this report means AI-feedback labeling for reward/preference construction. It is not full RLHF and it is not human preference learning.

11 Main Results

The project should be read as a sequence rather than as a single Phase 1D experiment: benchmark foundation, BudgetRAG context policies, adaptive allocation, RLAIF labels, reward/preference construction, offline selector evaluation, retriever diversity, and KV-cache motivation.

All results below are offline logged-candidate results. They are not human preference learning, not online RL, not DPO/PPO/GRPO, and not runtime KV-cache pruning. Context reward remains a non-default candidate.

11.1 Key Findings

Finding	Evidence	Interpretation
Full context-level RLAIF labels are available.	192/192 MiMo context labels were merged; 177 were clean usable labels; sufficiency rate was 0.591.	Context supervision is no longer only a skeleton or subset.
Retrieved contexts contain noise.	Mean selected chunks 1.276 versus mean irrelevant chunks 3.708.	Answer-level reward can hide weak evidence contexts.
Context reward changes the optimization target.	Penalty 0.25 changed 140/192 reward rows; preferences increased from 722 to 952; mean changed delta was -0.212.	Context reward is useful for calibration, but still non-default.
Pairwise RLAIF mostly agrees with scalar reward.	MiMo-50 pairwise audit agreement was 0.854 over non-ambiguous decisions; mean confidence was 0.914.	Scalar reward is not arbitrary, but needs calibration for near-tie answers.
Secondary judge supports the insufficiency signal.	DeepSeek v4 Flash agreed with MiMo on 80/83 comparable high-risk sufficiency decisions, agreement 0.964.	MiMo is not uniquely harsh on this targeted subset.
Retriever-diversity coverage is fixed.	Retrieval-only run produced 2250 actions across BM25, graph-BM25, and hybrid-RRF; A1-medium generation produced 1500/1500 non-empty rows with 0 errors and 0 empty answers.	Logged action coverage now supports retrieval-context allocation experiments.
A1 has answer-level and stratified context-level evidence.	A1 has 1500/1500 valid answer labels and 600 context labels with 548 clean usable rows. Graph-BM25 leads answer-level reward/quality, while hybrid-RRF leads context sufficiency/support on the subset.	Final retriever ranking remains metric-dependent and provisional.
A1 context-level evidence is now measured on a stratified subset.	The 600-row A1 context pass has sufficiency 0.6985, context quality 0.6431, evidence support 0.6567, and 3.487 irrelevant chunks on average.	This turns A1 from generation-only validation into retriever-diverse answer/context evaluation, but only over a stratified subset.
A1 answer and context signals diverge.	Graph-BM25 is strongest at answer level; hybrid-RRF has the highest context sufficiency/support in the 600-row subset, while graph-BM25 has lower token cost and higher KV savings.	Retriever ranking is metric-dependent and should remain query-conditioned.
A1 selector signal is positive but costly.	With context reward 0.25, `linear_reward_model` remains the strongest non-oracle selector: reward 0.669 ± 0.081, quality 0.967 ± 0.026, oracle gap 0.145 ± 0.082.	The learned selector still uses high-cost actions; context reward remains non-default.

11.2 Phase Timeline and Deliverables

Phase	Goal	Implementation	Key output/result
Phase 1A	Benchmark foundation	Retriever registry, benchmark CLI, retrieval metrics, and traceable records.	A reusable output schema for later BudgetRAG and RLAIF stages.

Phase 1B	Fixed BudgetRAG policies	Legacy, character budget, per-document budget, score-density, sentence-trim, and evidence-aware policies.	10-query SciFact smoke showed context compression and analytical KV savings can be measured independently from answer quality.
Phase 1B.1	Traceable matrix outputs	Matrix runner and summarizer with stable raw output directories.	Phase outputs became auditable and joinable for later RLAIF builders.
Phase 1C	Adaptive heuristic	Deterministic selector using score gap, entropy, confidence, and document-dominance features.	Initial adaptive path was stable but conservative.
Phase 1C.1	Larger retrieval validation	50-query SciFact BM25 retrieval-only validation.	Adaptive heuristic selected `evidence-aware` for 48 queries and `per-doc-budget` for 2, but still selected budget 4000 for all 50.
Phase 1C.2	Calibrated adaptive profiles	Conservative, balanced, and aggressive profiles with normalized gap/entropy thresholds.	Balanced/aggressive profiles created smaller-context stress tests; aggressive budget-4000 kept 2080 chars on average versus conservative 4000.
Phase 1C.3	Generation feedback	Multi-model generation, RAGAS join, MiMo answer labels, and long-context MiMo runs.	192 joined RLAIF action rows and 192 MiMo answer labels; MiMo quality vs RAGAS answer relevancy correlation 0.277.
Phase 1D	RLAIF reward/preference and offline selector	Answer/context/pairwise labelers, reward/preference builder, query-level split, selector baselines, and audits.	Full context labels, reward ablations, pairwise audit, action coverage diagnostics, and six-seed selector sweeps.
Retriever-diversity extension	Multi-retriever logged actions	BM25, graph-BM25, and hybrid-RRF retrieval-only and generation matrices.	Retrieval-only 2250-row coverage run; A1-medium 1500-row generation gate with complete answer labels and a 600-row stratified context-label pass.

11.3 Full Context-Level RLAIF Labels

The first full MiMo context-label pass merged all 192 Phase 1D selector-smoke action rows. Validation found no missing action ids, no unknown action ids, no duplicate action ids, no duplicate conflicts, and no invalid JSON labels.

Metric	Value
Merged context labels	192
Clean usable labels	177
Ambiguous labels	15
Invalid JSON labels	0
Sufficient contexts	110
Insufficient contexts	76
Sufficiency rate	0.591
Mean selected chunks	1.276
Mean redundant chunks	0.286
Mean irrelevant chunks	3.708
Mean context quality	0.602
Mean evidence support	0.556
Mean minimality	0.947

The important evidence is not only the sufficiency rate. It is the gap between selected and irrelevant chunks: the judge typically identifies a small evidence subset while marking several retrieved chunks as irrelevant. This supports the BudgetRAG/MemAlign direction: context selection should be supervised at the chunk level, not only judged after answer generation.

11.4 Context Reward Ablation

The context labels were merged into reward construction only through explicit, non-default candidates. Ambiguous context labels fall back to answer-level feedback. The answer-only reward remains the baseline.

Setting	Preferences	Changed rows	Negative deltas	Mean changed delta	Interpretation
answer-only	722	–	–	–	Baseline reward from answer labels/fallback feedback.
context penalty 0.25	952	140	108	-0.212	Conservative context candidate; useful for further audit.
context penalty 0.50	946	140	108	-0.316	Stronger penalty; lower absolute reward scale.
context penalty 1.00	944	140	108	-0.525	Too harsh for default use; diagnostic only.

The reward landscape changes materially: penalty 0.25 increases preference count from 722 to 952 and changes 140/192 reward rows. Most changes are negative because insufficient or noisy contexts are penalized. This is expected; it means the context labels are exposing evidence failures that answer-level labels alone do not represent.

11.5 Pairwise RLAIF Audit

Direct pairwise RLAIF was used to audit whether scalar reward-derived preferences align with an independent pairwise judge. In the MiMo-50 audit, Action A was the scalar reward-chosen action and Action B was the rejected action.

Metric	Value
Pair labels	50
Valid JSON	50
Ambiguous/error rows	2
A wins	41
B wins	7
Agreement over non-ambiguous decisions	0.854
Mean confidence	0.914
Small quality/support delta pairs	38
Cheaper wins when quality/support tied	35
Scalar-over-quality disagreements	5

The disagreement pattern is narrow: all seven direct-judge disagreements in the MiMo-50 audit came from `query_id=128`. In those cases the scalar reward favored slightly higher quality/support scores, while the pairwise judge considered both answers acceptable or both correct abstentions and then preferred the cheaper action. The calibration diagnostic therefore suggests a candidate tie band of quality 0.10 and support 0.20, but this remains an audit-derived candidate, not a default reward rule.

11.6 DeepSeek Secondary-Judge Audit

The retriever-diverse 300-row subset was used to select 100 high-risk context rows: MiMo-insufficient contexts, large negative context-reward deltas, high answer quality with low context support, and rows with many irrelevant chunks. DeepSeek v4 Flash was then used as a secondary context judge.

Metric	Value
Targeted rows	100
DeepSeek valid JSON labels	100
DeepSeek ambiguous labels	12
DeepSeek invalid JSON/errors	0
Comparable MiMo-vs-DeepSeek sufficiency rows	83
Agreement	80/83 = 0.964
Consensus-insufficient rows	76
High-disagreement rows	3
MiMo-harsh rows	1
Evidence-support Pearson correlation	0.905
Context-quality Pearson correlation	0.476

This audit does not turn AI feedback into ground truth. It does, however, reduce the concern that MiMo is uniquely harsh on high-risk rows. The stronger agreement on evidence support than on raw context quality also suggests that sufficiency/support labels are safer reward inputs than an uncalibrated context-quality scalar.

11.7 Offline Selector Results

The selector is evaluated only over logged candidate actions. The artifact keeps `runtime_default_replacement=false`; it does not replace the runtime `adaptive-heuristic` policy.

Policy	Reward	Quality	Coverage	Oracle gap
cheapest	0.577 ± 0.159	0.770 ± 0.152	1.000 ± 0.000	0.094
best average	0.618 ± 0.147	0.794 ± 0.125	0.911 ± 0.050	0.080
shrinkage smoothed best average	0.602 ± 0.117	0.773 ± 0.097	1.000 ± 0.000	0.070
smoothed linear selector	0.595 ± 0.137	0.767 ± 0.122	1.000 ± 0.000	0.076
oracle logged	0.671 ± 0.134	0.834 ± 0.138	1.000 ± 0.000	0.000

Takeaway: `shrinkage_smoothed_best_average` is the best full-coverage non-oracle baseline by reward and oracle gap, while `best_average` still has higher reward/quality on the query groups it covers. This indicates that action representation and logged-action coverage remain stronger bottlenecks than adding a more complex RL algorithm. Learned linear selectors are functional and full-coverage, but they do not yet robustly beat simple smoothed baselines.

11.8 Retriever-Diversity Extension

The first Phase 1D logs were effectively BM25-only. The retriever-diversity extension fixes this logged-action coverage gap, then starts adding quality supervision.

Run	Actions	Retrievers	Scored rewards	Purpose / conclusion
Retrieval-only limit50	2250	3	0	Coverage run only: every sampled query has BM25, graph-BM25, and hybrid-RRF rows across 45 action configurations.
300-row generation subset	300	3	186	Diagnostic quality run: graph-BM25 had slightly better answer support/context quality; hybrid-RRF was noisier, but `MAX_COMPLETION_TOKENS=256` caused 77 empty answers.
A1-medium generation + answer labels	1500	3	1460	Generation gate plus answer-level RLAIF: 1500/1500 valid MiMo V2.5 answer labels, 1460 clean usable labels, 0 invalid JSON, and 17026 answer-only preference pairs.
A1 stratified context labels + reward	600 labels / 1500 rewards	3	1492	Context-level RLAIF subset: 548 clean usable context labels, sufficiency 0.6985, and a non-default context reward candidate changing 464 reward rows.

A1 answer-label validation metric	Value
Action rows	1500
Valid MiMo V2.5 labels	1500
Clean usable labels	1460
Ambiguous labels	40
Invalid JSON labels	0
Missing / unknown / duplicate action ids	0
Scored answer-only rewards	1460
Answer-only preferences	17026

Retriever	Rows	Quality	Support	Unsupported risk	Reward	KV cost
BM25	500	0.899	0.899	0.096	0.617	0.605
graph-BM25	500	0.911	0.909	0.087	0.642	0.559
hybrid-RRF	500	0.862	0.862	0.141	0.546	0.673

Retriever	Clean rows	Suff.	Context quality	Evidence support	Token cost
BM25	182	0.720	0.647	0.685	687.247
graph-BM25	180	0.706	0.642	0.659	579.606
hybrid-RRF	186	0.780	0.682	0.730	689.935

This turns A1 from a generation-only validation into a retriever-diverse answer/context evaluation, but only over a stratified 600/1500 context-label subset.

A1 answer-only selector	Cov.	Reward	Quality	Token	KV	Gap
cheapest	0.900 ± 0.058	0.270 ± 0.115	0.702 ± 0.056	0.249 ± 0.002	0.248 ± 0.000	0.546 ± 0.116
best avg.	1.000 ± 0.000	0.661 ± 0.111	0.937 ± 0.070	0.554 ± 0.135	0.758 ± 0.260	0.154 ± 0.111
shrinkage avg.	1.000 ± 0.000	0.659 ± 0.112	0.938 ± 0.072	0.577 ± 0.137	0.808 ± 0.265	0.156 ± 0.111
linear model	1.000 ± 0.000	0.734 ± 0.032	0.987 ± 0.018	0.656 ± 0.014	0.994 ± 0.001	0.081 ± 0.032
smoothed linear	0.967 ± 0.047	0.657 ± 0.126	0.931 ± 0.076	0.480 ± 0.121	0.659 ± 0.240	0.158 ± 0.125
oracle	1.000 ± 0.000	0.815 ± 0.001	1.000 ± 0.000	0.264 ± 0.008	0.249 ± 0.000	0.000 ± 0.000

A1 context-label validation metric	Value
Context labels	600
Valid JSON labels	598
Clean usable labels	548
Ambiguous labels	52
Invalid JSON labels	2
Missing / unknown / duplicate action ids	0
Sufficiency rate	0.6985
Mean selected chunks	1.2300
Mean irrelevant chunks	3.4867
Mean context quality	0.6431
Mean evidence support	0.6567

A1 context-candidate selector	Cov.	Reward	Quality	Token	KV	Gap
cheapest	1.000 ± 0.000	0.358 ± 0.104	0.793 ± 0.041	0.249 ± 0.002	0.248 ± 0.000	0.456 ± 0.105
best avg.	0.983 ± 0.037	0.613 ± 0.127	0.906 ± 0.061	0.321 ± 0.052	0.298 ± 0.082	0.201 ± 0.126
shrinkage avg.	0.983 ± 0.037	0.555 ± 0.096	0.885 ± 0.045	0.320 ± 0.053	0.310 ± 0.079	0.259 ± 0.096
linear model	1.000 ± 0.000	0.669 ± 0.081	0.967 ± 0.026	0.623 ± 0.017	0.982 ± 0.028	0.145 ± 0.082
smoothed linear	0.983 ± 0.037	0.619 ± 0.133	0.925 ± 0.048	0.379 ± 0.083	0.473 ± 0.189	0.195 ± 0.131
oracle	1.000 ± 0.000	0.814 ± 0.003	1.000 ± 0.000	0.266 ± 0.014	0.273 ± 0.035	0.000 ± 0.000

Context-candidate selector	BM25	graph-BM25	hybrid-RRF
best avg.	42	17	1
shrinkage avg.	31	27	2
linear model	8	51	1
smoothed linear	38	19	3
oracle	31	20	9

Takeaway: the retriever-diversity extension has fixed the logged-action coverage problem and now has answer-level and stratified context-level quality labels. Graph-BM25 is the strongest mean answer-level retriever in A1. The 600-row context labels do not simply confirm that ranking: hybrid-RRF has the best context sufficiency/support, while graph-BM25 has lower token cost and more KV savings. With context reward 0.25, `linear_reward_model` is still the strongest non-oracle selector by reward and oracle gap, but it favors graph-BM25 and high KV-cost actions. The remaining caveat is calibration and coverage:

> Retriever diversity coverage, answer-level A1 evaluation, and a 600-row stratified context-label pass are complete. A final BM25 vs graph-BM25 vs hybrid-RRF ranking remains provisional because context labels cover 600/1500 actions and answer-level/context-level signals disagree.

12 Qualitative Error Analysis

The aggregate metrics are best interpreted together with individual cases. This section summarizes representative rows from the local experiment artifacts; raw JSONL outputs are not included in the report repository.

12.1 Correct Abstention but Insufficient Context

Several rows receive high answer-level quality because the model correctly refuses or abstains, while the context-level label still marks the retrieval context as insufficient. The two labels measure different objects: *answer behavior* and *retrieved evidence quality*.

Query	Claim	Answer pattern	Context label
75	Active *H. pylori* urease has a polymeric structure comprising UreA and UreB.	The answer says the provided contexts do not contain the urease-structure information. MiMo answer quality/support are 1.0.	Insufficient; no selected chunks; 9 irrelevant chunks; context quality 0.0.
72	Activator-inhibitor pairs are provided dorsally by Admpchordin.	The answer says the contexts do not mention Admpchordin or dorsal activator-inhibitor pairs. Answer quality/support are 1.0.	Insufficient; no selected chunks; 8 irrelevant chunks; context quality 0.0.

These examples motivate context-level RLAIF. Answer labels alone would treat the abstention as correct, but the retriever/context layer still failed to provide evidence.

12.2 Sufficient Context with Many Irrelevant Chunks

Other rows are sufficient because one chunk contains the needed evidence, even though most retrieved chunks are irrelevant. This pattern motivates evidence-aware context allocation and future KV evidence masks.

Query	Claim	Useful evidence pattern	Irrelevant chunks
49	ADAR1 binds to Dicer to cleave pre-miRNA.	One selected chunk directly states that ADAR1 binds Dicer and promotes pre-miRNA cleavage.	8
48	A total of 1,000 people in the UK are asymptomatic carriers of vCJD infection.	One selected chunk provides prevalence data contradicting the 1,000 claim.	7
42	High microerythrocyte count raises vulnerability to severe anemia in homozygous alpha-thalassemia trait subjects.	One selected chunk shows the count is protective against severe malarial anemia, not harmful.	7

This pattern explains why the average selected chunk count is only 1.276 while the average irrelevant chunk count is 3.708. It also shows why context reduction can improve efficiency without necessarily removing evidence, provided that the policy can identify the useful chunk.

12.3 Pairwise Disagreement: Quality Tie, Efficiency Wins

The MiMo-50 pairwise audit found 7 disagreements between scalar reward-derived preferences and direct pairwise judge choices. They cluster in query 128. The scalar reward preferred a slightly higher quality/support action; the pairwise judge considered both answers acceptable, or both abstentions correct, and chose the cheaper action. This motivates `pairwise_tie_v1`: when quality/support differences fall within a small tie band, resource efficiency should carry more weight.

12.4 MiMo-Harsh / High-Disagreement Rows

The multi-judge audit identified 6 rows where MiMo marked context insufficient while at least one secondary judge marked it sufficient. These rows are calibration examples. They should be reviewed before increasing context penalties or treating MiMo context labels as clean evidence-mask supervision.

13 Offline Selector / Bandit-Style Baselines

The selector baselines choose among logged candidates only. They do not replace the runtime adaptive heuristic. The main policies are:

- `fixed`: choose one frequent signature; low coverage.
- `cheapest`: choose the lowest resource-cost action.
- `best_average`: choose the exact action signature with highest train mean reward.
- `family_smoothed_best_average`: back off from exact signature to retrieval-context family to context policy.
- `shrinkage_smoothed_best_average`: shrink exact/family/context means toward broader means.
- `linear_reward_model`: ridge/linear offline selector over non-leaking action and cost features.

- `smoothed_linear_selector`: add train-only aggregate means/counts to the linear selector.
- `oracle_logged`: logged-candidate upper bound, not deployable.

The six-seed answer-only sweep showed that `best_average` has the highest non-oracle reward when it covers a group, but coverage is 0.911. `shrinkage_smoothed_best_average` is the strongest full-coverage non-oracle selector so far: reward 0.602, quality 0.773, oracle gap 0.070. The learned linear selectors are functional but not yet robustly stronger than simple smoothed baselines.

Policy	Coverage	Reward	Quality	Oracle gap
cheapest	1.000	0.577	0.770	0.094
best average	0.911	0.618	0.794	0.080
shrinkage smoothed	1.000	0.602	0.773	0.070
linear reward model	1.000	0.586	0.774	0.086
smoothed linear	1.000	0.595	0.767	0.076
oracle logged	1.000	0.671	0.834	0.000

The current bottleneck is data and action coverage, not missing algorithmic complexity.

14 KV-Cache Motivation

The analytical KV-cache estimate used in the project is:

$$\text{KV bytes} \approx 2 \times L \times H_{kv} \times D_{head} \times S \times B \times \text{bytes per value.}$$

For GQA/MQA models, H_{kv} should be the number of key-value heads, not the number of attention heads. The estimates exclude model weights, activations, allocator overhead, paged-attention block overhead, and runtime workspaces.

Model	KV GiB @ 4K	KV GiB @ 32K	KV GiB @ 128K
Qwen2.5-0.5B	0.047	0.375	1.500
Qwen2.5-1.5B	0.109	0.875	3.500
Qwen2.5-3B	0.141	1.125	4.500
Qwen2.5-7B	0.219	1.750	7.000
Qwen2.5-14B	0.750	6.000	24.000

BudgetRAG is the pre-inference step: reduce or improve context before the model runs. Future MemAlign-Qwen work can compare full versus compressed context on local Qwen models, then test runtime KV retention using sink, recent, and evidence tokens.

15 Limitations

- The labels are AI feedback, not human annotations.
- The selector is offline logged-candidate evaluation, not online RL.
- Current logged action coverage is still small and retriever diversity is low.
- Current results support context-budget/action selection more strongly than retrieval-strategy selection.

- Context reward is non-default.
- Pairwise RLAIF currently audits and calibrates scalar reward; it does not train a pairwise selector yet.
- No DPO, PPO, GRPO, online RL, or production runtime KV pruning is implemented.
- Web-search actions, if used, are live stress tests and must not be mixed with reproducible BEIR-style benchmark claims.
- Future MiMo 2.5 runs use standard `mimo-v2.5`; previous Pro-labeled rows remain historical experiment provenance. Merged tables may include both, but they must annotate judge-model drift and keep model provenance visible.

16 Future Work

The next steps are:

1. Inspect the 6 MiMo-harsh rows and consensus-insufficient examples.
2. Log retrieval-diversity runs with BM25, graph-BM25, and hybrid-RRF, then analyze context-label evidence quality by retriever and policy.
3. Increase query groups to reduce split variance.
4. Train context-label-aware selectors once supervision is larger.
5. Add a pairwise ranker after enough direct pair labels exist.
6. Profile local Qwen full versus compressed contexts, including prefill/decode latency and peak memory.
7. Later, test a runtime KV-pruning proof of concept that retains sink, recent, and evidence tokens.

17 Threats to Validity

17.1 AI-Judge Bias

MiMo, DeepSeek, Groq, and RAGAS are not human annotators. They may disagree on sufficiency, support, abstention quality, or unsupported-claim risk. The multi-judge audit reduces dependence on a single judge, but it does not replace human validation.

17.2 Logged-Candidate Bias

Offline selectors can only choose among actions that were logged. If the matrix lacks a useful retriever, budget, policy, or model, the selector cannot discover it. This is why retriever-diversity logging is more important than adding complex RL at the current stage.

17.3 Small Held-Out Splits

The split rule avoids query leakage, but the eval sets remain small. Seed 42 looked optimistic for some learned selectors; six-seed evaluation gave a more realistic view. Current selector conclusions should be interpreted as early diagnostic signals.

17.4 RAGAS vs MiMo Scale Mismatch

MiMo quality and RAGAS answer relevancy have correlation 0.277. Absolute reward values from RAGAS-only and MiMo-judge runs should not be compared as if they share the same scale. The useful comparison is the policy ordering and the direction of trade-offs.

17.5 Context Reward Calibration

Context labels materially change reward rows and preference counts. However, penalty 1.00 is too aggressive for a default. Penalty 0.25 is a conservative candidate, but the MiMo-harsh rows need qualitative review before context reward is treated as stable supervision.

18 Completed and Planned Evaluations

18.1 HotpotQA Sampled Kaggle Evaluation

HotpotQA is now included as a sampled Kaggle evaluation, not a full benchmark claim. It is valuable because multi-hop QA stresses evidence sufficiency more directly than simple claim verification. The implementation uses cached BM25 retrieval so the 5.23M-document BEIR HotpotQA corpus is indexed once per notebook instead of once per action row.

The clean MiMo sampled runs provide the usable HotpotQA evidence in this report:

- **Full 16-action sampled matrix:** 800/800 MiMo generations succeeded on 50 sampled queries, with 80 MiMo-backed RAGAS samples. Cached BM25 retrieval gives mean recall@10 0.610 and nDCG@10 0.583. The best token-F1 row is `adaptive-heuristic__balanced__16000` at 0.097, while the best RAGAS answer-relevancy row is `adaptive-heuristic__aggressive__4000` at 0.871.
- **Evidence-aware budget curve:** 300/300 generations succeeded for 1k through 32k budgets, with 30 RAGAS samples. Evidence-aware context saturates after 8k in this sample; 8k is the most defensible trade-off row because it improves answer relevancy without retaining more context at 16k/32k.
- **Low-budget curve:** legacy and adaptive rows over 500–3000 characters completed 750/750 generations, with 75 RAGAS samples. This run tests whether the adaptive policy remains usable under aggressive context limits.

Exact match is 0.000 across the main sampled matrix, so token-F1 and judge metrics are the relevant answer-quality signals for this stage. The Groq Qwen3-32B HotpotQA run is kept as a diagnostic because provider rate limits caused 417/800 generation errors; it must not be mixed with the clean MiMo summary.

18.2 vLLM / Vast AI Model-Bench Workflow

The merged model-bench workflow is an operator/profiling path, not a selector-quality benchmark. It is included because deployment planning for BudgetRAG and future local Qwen experiments depends on knowing whether candidate models can be served reliably on rented GPUs. The workflow benchmarks local or existing OpenAI-compatible vLLM endpoints, records latency, time-to-first-token, tokens per second, hardware samples, and serving configuration fields such as quantization, KV-cache dtype, attention backend, speculative decoding settings, eager/CUDA-graph mode, batch limits, cache limits, and CPU offload.

The current Vast AI scripts target RTX 5060 Ti 16GB hosts with CUDA 13.0 and CUDA 12.9 profiles. Their outputs are written under ignored `runs/model_bench/` directories. These measurements should be used for deployment triage and cost planning; they are not mixed with the RLAIF selector, HotpotQA, or BEIR benchmark claims.

18.3 Retriever Diversity

Current evidence mostly supports context-budget/action selection. To claim retrieval-context allocation, logged actions must include multiple retrieval strategies. A first retrieval-only SciFact run covers 45 jobs over 50 queries:

- 3 retrievers: BM25, graph-BM25, and hybrid-RRF;
- 5 policy/profile variants over 3 budgets, producing 2250 normalized action rows;
- every query has all 3 retrievers and all 45 action rows;
- generation was skipped, so this verifies logged action coverage but does not provide answer-quality or judge-label evidence.

The follow-up step was a small MiMo-v2.5 generation subset with explicit `retriever`, `context_policy`, `adaptive_profile`, `budget`, `generator_model`, and `judge_model` columns. Optional web-search remains a live stress test only and must not be mixed with reproducible BEIR claims. This generation subset has now been run for 10 SciFact queries:

- 30 jobs produced 300 retriever-diverse action rows over BM25, graph-BM25, and hybrid-RRF with standard `mimo-v2.5`.
- MiMo answer judging produced 300 labels, 299 valid JSON labels, 222 labels with numeric diagnostics, and 186 clean reward rows after filtering ambiguous labels.
- The reward build produced 1559 preference pairs, including 1189 retrieval-context preferences.
- Separate validation and answer-quality diagnostics now report generation coverage, missing-answer rate, answer quality, support, unsupported-claim risk, and cost by retriever and context policy.
- Full context-level MiMo judging now covers the same 300 action rows: 300 valid JSON context labels, 253 clean usable labels, 134 sufficient contexts, 158 insufficient contexts, mean context quality 0.505, and mean evidence support 0.436.
- A non-default context reward candidate changed 156/300 reward rows, mostly downward, and increased preference pairs from 1559 to 2412. This confirms that context labels add evidence-quality supervision, but the candidate remains a calibration target rather than a default selector target.
- Six-seed held-out selector evaluation is only a diagnostic because the run contains 10 query groups; it verifies the pipeline but does not support a learned-selector generalization claim.

The run also exposed an operational issue: `MAX_COMPLETION_TOKENS=256` is too low for standard MiMo V2.5 generation, producing 77 empty answer strings with no request errors. Future retriever-diverse generation runs should use a larger generation cap before scaling to the full 50-query matrix.

The next-run decision therefore keeps the retriever-diverse branch active but avoids the full 2250-row matrix for now. The recommended follow-up was a 50-query, two-budget run with standard MiMo V2.5 and `MAX_COMPLETION_TOKENS=2048`, plus a targeted DeepSeek audit over 100 high-impact context rows. Local Qwen profiling is deferred from this branch.

The targeted DeepSeek audit is complete: DeepSeek v4 Flash produced 100 valid context labels with no invalid JSON or request errors, and MiMo-vs-DeepSeek sufficiency agreement is 80/83 over comparable rows. The audit found 76 consensus-insufficient rows and only 3 high-disagreement rows, supporting the direction of the MiMo context-insufficiency signal while keeping context reward non-default.

The A1-medium generation, answer-label, and stratified context-label gates are now complete. The run covers 50 SciFact queries, BM25, graph-BM25, hybrid-RRF, five policy/profile variants, budgets `1000` and `4000`, and standard `mimo-v2.5` with `MAX_COMPLETION_TOKENS=2048`. It produced 1500/1500 non-empty generated answers, 0 generation errors, 30 query-result files, and 1500 normalized RLAIF action rows. The three Kaggle answer-label shards merged into 1500/1500 valid MiMo V2.5 answer labels, with 1460 clean usable labels, 40 ambiguous labels, 0 invalid JSON lines, no missing action ids, no unknown action ids, and no duplicate conflicts.

The answer-only reward rebuild produced 1500 reward rows, 1460 scored rewards, and 17026 preference pairs. The stratified 600-row context-label pass produced 548 clean usable labels, 52 ambiguous labels, 2 invalid JSON labels, no missing/unknown/duplicate action ids, sufficiency 0.6985, mean context quality 0.6431, and mean evidence support 0.6567. A non-default context reward candidate with penalty 0.25 produced 1500 reward rows, 1492 scored rewards, and 20934 preferences, changing 464 rewards relative to answer-only. The answer-level retriever signal favors graph-BM25, while the context-level subset favors hybrid-RRF on sufficiency/support and graph-BM25 on cost/KV savings. The retriever-quality ranking is therefore provisional and metric-dependent, not a final global ranking.

18.4 Larger Logged Query Groups

The action coverage diagnostics show exact-signature sparsity. More query groups and broader logged action families are needed before a learned selector can robustly beat strong smoothed baselines.

19 Implementation Map

Component	Purpose	Main output
CLI layer	Entry points for benchmark, BudgetRAG, and RLAIF commands.	User-facing commands.
Action normalization	Normalize retrieval-context action dimensions.	Stable action signatures.
RLAIF schemas	Schemas for actions, feedback, rewards, and preferences.	Validated JSONL records.
Dataset builder	Convert BudgetRAG `query_results.jsonl` into normalized datasets.	actions and feedback JSONL.
Answer labeler	AI-judge answer labels with resume and JSON repair.	answer-label JSONL.
Answer-label validator	Merge and validate sharded answer-label outputs.	merged labels and validation summary.
Context labeler	AI-judge context sufficiency and chunk labels.	context-label JSONL.
Stratified action sampler	Select balanced context-label subsets by retriever, policy/profile, and budget.	sampled action JSONL and summary.
Reward builder	Build scalar rewards and preferences with guardrails.	rewards and preferences JSONL.

Query splitter	Deterministic query-level split.	train/eval JSONL and manifest.
Selector policies	Offline selector baselines.	policy artifact and eval summary.
Pairwise audit	Direct pairwise judge audit.	pairwise-label JSONL.
Context ablation runner	Full postprocess pipeline for context labels.	validation, reward deltas, selector sweeps.
Multi-judge aggregator	Aggregate MiMo/DeepSeek/Groq audit labels.	multi-judge report and summary JSON.
Evidence-quality analyzer	Summarize context-label evidence quality by retriever and context policy.	policy-level evidence diagnostics.
KV estimator	Analytical Qwen KV-cache estimates.	KV estimate report.
Model-bench workflow	Benchmark local or existing vLLM endpoints and record serving and hardware metadata.	model-bench manifests, metrics, and summaries.

The corresponding source files are under `src/rag_bench/` for modules and `scripts/` for standalone analysis scripts.

The code-level test suite covers schema validation, builders, reward/preference logic, labeler hardening, split behavior, selector policies, analysis scripts, model-bench helpers, and secret handling. The merged internship branch currently verifies with 271 passing tests.

A Appendix A: Phase 1B–1C Tables

A.1 Phase 1B Smoke Matrix

Policy / budget	Queries	Kept chars	Compression	Token savings	KV savings
char-budget 1000	10	999.7	0.2161	961.0	900.9
char-budget 2000	10	2000.0	0.4324	711.0	666.6
char-budget 4000	10	3906.0	0.8380	234.2	219.6
evidence-aware 1000	10	999.7	0.2162	961.0	900.9
evidence-aware 2000	10	2000.0	0.4324	711.0	666.6
evidence-aware 4000	10	3999.0	0.8647	234.3	219.7
legacy 1000	10	1002.0	0.2167	960.4	900.4
legacy 2000	10	2005.0	0.4336	709.6	665.2
legacy 4000	10	3914.0	0.8396	232.0	217.5
score-density 1000	10	999.7	0.2162	961.0	900.9
score-density 2000	10	2000.0	0.4324	711.0	666.6
score-density 4000	10	3906.0	0.8380	234.2	219.6

A.2 Phase 1C.2 Adaptive Profile Matrix

Profile / budget	Queries	Kept chars	Compression	Token savings	KV savings
aggressive 1000	50	999.8	0.1248	1829	1715
aggressive 2000	50	1360	0.1665	1739	1630
aggressive 4000	50	2080	0.2500	1559	1462
balanced 1000	50	2260	0.2816	1514	1419

balanced 2000	50	2840	0.3540	1369	1283
balanced 4000	50	4000	0.4991	1079	1012
conservative 1000	50	4000	0.4991	1079	1012
conservative 2000	50	4000	0.4991	1079	1012
conservative 4000	50	4000	0.4991	1079	1012
evidence-aware 1000	50	999.8	0.1248	1829	1715
evidence-aware 2000	50	2000	0.2495	1579	1480
evidence-aware 4000	50	4000	0.4991	1079	1012
legacy 1000	50	1002	0.1250	1829	1714
legacy 2000	50	2006	0.2503	1577	1479
legacy 4000	50	4014	0.5009	1075	1008

Balanced-profile diagnostics at matrix budget 1000: normalized score gap min 0.0071, mean 0.2012, max 0.5288; normalized score entropy min 0.9401, mean 0.9861, max 0.9999; score confidence min 0.00000037, mean 0.0043, max 0.0236.

B Appendix B: RLAIF Dataset and Label Tables

B.1 Selector Smoke Build Summary

Metric	Count
RAGAS source action files	64
Completed RAGAS samples	192
Joined action-query rows	192
`rlaif_actions.jsonl` rows	192
`rlaif_feedback.jsonl` rows	192
`rlaif_rewards.jsonl` rows	192
`rlaif_preferences.jsonl` rows	578
Query groups	55
Action signatures	77
Invalid rows	0
Context-policy preferences	289
Retrieval-context preferences	289
Skipped small reward delta	694
Skipped quality guardrail failed	2

B.2 MiMo Answer Label Summary

Metric	Value
Label count	192
Valid JSON	192
Invalid JSON	0
Ambiguous labels	25
Judge errors	1
Scored labels	191
Used MiMo labels in reward	167
Fallback to original feedback	25

MiMo quality vs RAGAS correlation	0.277

Score	N	Mean	Std	Min	Max
`overall_quality`	191	0.823	0.307	0.000	1.000
`answer_correctness`	191	0.857	0.313	0.000	1.000
`evidence_support`	191	0.825	0.340	0.000	1.000
`faithfulness`	189	0.847	0.317	0.000	1.000
`unsupported_claim_penalty`	191	0.090	0.280	0.000	1.000

B.3 Full Context Label Validation

Shard	Rows	Clean usable
1–50	50	46
51–121	71	66
122–192	71	65
Total	192	177

Validation metric	Value
Action count	192
Merged labels	192
Missing actions	0
Unknown action ids	0
Duplicate action ids	0
Duplicate conflicts	0
Clean usable labels	177
Ambiguous labels	15
Invalid JSON labels	0
Dropped unknown chunk ids	0

B.4 Context Reward Delta Summary

Penalty	Changed	Negative	Positive	Mean all delta	Mean changed delta
0.25	140	108	32	-0.155	-0.212
0.50	140	108	32	-0.231	-0.316
1.00	140	108	32	-0.383	-0.525

Penalty	Sufficient false	Sufficient true	Missing sufficiency
0.25	64	72	4
0.50	64	72	4
1.00	64	72	4

B.5 Action Coverage Diagnostics

Level	Unique	Singleton rate	Eval group coverage
`action_id`	192	1.000	0.000
`exact_signature`	77	0.260	0.911
`retrieval_context_family`	16	0.000	1.000
`context_policy`	3	0.000	1.000
`retriever`	1	0.000	1.000

C Appendix C: Selector, Pairwise, and Multi-Judge Tables

C.1 Answer-Only Six-Seed Selector Sweep

Policy	Coverage	Reward	Quality	Oracle gap
fixed	0.026 ± 0.057	0.829 ± 0.000	1.000 ± 0.000	0.007
cheapest	1.000 ± 0.000	0.577 ± 0.159	0.770 ± 0.152	0.094
best average	0.911 ± 0.050	0.618 ± 0.147	0.794 ± 0.125	0.080
family smoothed	1.000 ± 0.000	0.598 ± 0.138	0.767 ± 0.122	0.074
shrinkage smoothed	1.000 ± 0.000	0.602 ± 0.117	0.773 ± 0.097	0.070
linear reward model	1.000 ± 0.000	0.586 ± 0.153	0.774 ± 0.110	0.086
smoothed linear	1.000 ± 0.000	0.595 ± 0.137	0.767 ± 0.122	0.076
oracle logged	1.000 ± 0.000	0.671 ± 0.134	0.834 ± 0.138	0.000

C.2 Context-Reward Candidate Selector Sweep

Candidate	Policy	Coverage	Reward	Quality	Oracle gap
penalty 0.25	cheapest	1.000	0.451	0.704	0.093
penalty 0.25	best average	0.911	0.478	0.701	0.085
penalty 0.25	shrinkage	1.000	0.449	0.683	0.095
penalty 0.25	smoothed linear	1.000	0.472	0.689	0.072
penalty 0.25	oracle	1.000	0.544	0.755	0.000
penalty 0.50	cheapest	1.000	0.386	0.704	0.104
penalty 0.50	best average	0.911	0.420	0.701	0.091
penalty 0.50	shrinkage	1.000	0.389	0.683	0.100
penalty 0.50	smoothed linear	1.000	0.415	0.689	0.074
penalty 0.50	oracle	1.000	0.489	0.755	0.000
penalty 1.00	cheapest	1.000	0.255	0.704	0.132
penalty 1.00	best average	0.911	0.303	0.701	0.109
penalty 1.00	shrinkage	1.000	0.273	0.689	0.114
penalty 1.00	smoothed linear	1.000	0.329	0.703	0.058
penalty 1.00	oracle	1.000	0.387	0.748	0.000

C.3 Direct Pairwise MiMo-50 Audit

Metric	Value
Pair labels	50
Valid JSON	50
Invalid JSON	0
Ambiguous / timeout rows	2
Separate judge/API errors	0
A wins	41
B wins	7
Agreement over non-ambiguous	0.854
Mean confidence	0.914
Small quality/support delta pairs	38
Cheaper wins when tied	35
Scalar-over-quality disagreements	5

C.4 DeepSeek Secondary-Judge Audit

Judge	Labels	Valid	Ambiguous	Clean sufficiency
MiMo	100	100	3	95
DeepSeek	100	100	12	88

Pair	Compared	Agree	Agreement rate
MiMo vs DeepSeek	83	80	0.964

Audit signal	Count
Consensus insufficient cases	76
Majority vote insufficient	93
Majority vote sufficient	4
Majority vote tie	3
High-disagreement cases	3
MiMo-harsh cases	1

Score comparison	N	Pearson
MiMo vs DeepSeek context quality	85	0.476
MiMo vs DeepSeek evidence support	85	0.905

D Appendix D: KV Estimates, CLI Chronology, and Sources

D.1 Full Qwen KV-Cache Estimate Table

Table D1 keeps the full analytical KV-cache evidence, but presents it as a compact matrix rather than one row per model/sequence pair. Values are GiB for batch size 1 and fp16/bf16 KV values; they exclude weights, activations, allocator overhead, and paged-attention block overhead.

Model	L/KV	1K	2K	4K	8K	16K	32K	128K
Qwen2.5-0.5B	24/2	0.012	0.023	0.047	0.094	0.188	0.375	1.500
Qwen2.5-1.5B	28/2	0.027	0.055	0.109	0.219	0.438	0.875	3.500
Qwen2.5-3B	36/2	0.035	0.070	0.141	0.281	0.562	1.125	4.500
Qwen2.5-7B	28/4	0.055	0.109	0.219	0.438	0.875	1.750	7.000
Qwen2.5-14B	48/8	0.188	0.375	0.750	1.500	3.000	6.000	24.000

The main implication is the linear growth with sequence length: for Qwen2.5-14B, moving from 4K to 32K raises the estimated KV cache from 0.75 GiB to 6.00 GiB before any runtime overhead is counted. This justifies treating context allocation as a first-class resource decision.

D.2 CLI Chronology

```
rag-bench run / scripts/run_budgetrag_matrix.py
scripts/summarize_budgetrag_results.py
rag-bench rlaif-build
rag-bench rlaif-label-answers
rag-bench rlaif-label-contexts
rag-bench rlaif-reward
rag-bench rlaif-split
rag-bench rlaif-train
rag-bench rlaif-eval
rag-bench rlaif-label-pairs
scripts/summarize_rlaif_labels.py
scripts/summarize_rlaif_context_labels.py
scripts/validate_rlaif_context_labels.py
scripts/run_context_reward_ablation_pipeline.py
scripts/aggregate_rlaif_multijudge_audit.py
```

D.3 Committed Source Reports Used

- docs/reports/phase1b_smoke_results.md
- docs/reports/phase1c_adaptive_smoke_results.md
- docs/reports/phase1c1_adaptive_validation.md
- docs/reports/phase1c2_adaptive_threshold_calibration.md
- docs/reports/phase1d_rlaif_selector_smoke.md
- docs/reports/phase1d_rlaif_heldout_eval.md
- docs/reports/phase1d_rlaif_ai_judge_heldout_eval.md
- docs/reports/phase1d_rlaif_pairwise_mimo50.md
- docs/reports/phase1d_rlaif_full_context_reward_ablation.md
- docs/reports/phase1d_rlaif_multijudge_audit.md
- docs/reports/local_qwen_kv_estimates.md

E Appendix E: From Experiment Logs to Final Report

This appendix maps the committed Markdown experiment reports to the PDF sections where their results are summarized. It is included to make the consolidation path explicit and auditable.

Markdown report	PDF location	Main use
docs/reports/phase1b_smoke_results.md	Phase 1B; Appendix A	Fixed-policy smoke matrix and compression/KV context.
docs/reports/phase1c_adaptive_smoke_results.md	Phase 1C; timeline	Initial adaptive heuristic behavior.
docs/reports/phase1c1_adaptive_validation.md	Phase 1C; timeline	50-query adaptive validation and conservative-budget observation.
docs/reports/phase1c2_adaptive_threshold_calibration.md	Phase 1C; Appendix A	Adaptive profile calibration.
docs/reports/phase1c3_multi_model_generation.md	Phase 1C.3	Multi-model generation coverage, latency, and provider-limit caveats.
docs/reports/phase1c3_mimo_long_context.md	Phase 1C.3; KV motivation	MiMo long-context budget behavior and analytical KV savings.
docs/reports/phase1d_rlaif_selector_smoke.md	Phase 1C.3; Appendix B	192 joined rows, RAGAS feedback, initial selector-smoke counts.
docs/reports/phase1d_rlaif_full_context_reward_ablation.md	Main Results; Appendix B/C	Full 192-row context labels, reward ablation, selector sweeps.
docs/reports/phase1d_rlaif_context_reward_candidate.md	Main Results	MiMo50 context-candidate setup and non-default reward framing.
docs/reports/phase1d_rlaif_v2_multiseed_selector_eval.md	Main Results; Appendix C	Six-seed answer-only selector metrics and limitations.
docs/reports/phase1d_rlaif_v2_shrinkage_selector_eval.md	Main Results; Appendix C	Shrinkage-smoothed selector result and full-coverage interpretation.
docs/reports/phase1d_rlaif_pairwise_mimo50.md	Main Results; Appendix C	Direct pairwise agreement and confidence.
docs/reports/phase1d_rlaif_pairwise_calibration_diagnostics.md	Main Results; Appendix C	Near-tie quality/support calibration pattern.
docs/reports/phase1d_rlaif_action_coverage.md	Main Results; Appendix B	Exact-signature sparsity and family-level coverage.
docs/reports/phase1d_retriever_diversity_retrieval_only_limit50.md	Main Results; completed evaluations	2250-action retrieval-only coverage run.
docs/reports/phase1d_retriever_diversity_generation_mimo10.md	Main Results; completed evaluations	300-row retriever-diverse generation, answer/context labels, and cap-256 empty-answer issue.
docs/reports/phase1d_retriever_diversity_deepseek_audit.md	Main Results; Appendix C	DeepSeek secondary-judge agreement on high-risk context rows.
docs/reports/phase1d_retriever_diversity_next_run_decision.md	Main Results; completed evaluations	A1-medium decision and guardrails.
docs/reports/phase1d_retriever_diversity_a1_medium_generation_validation.md	Main Results; completed evaluations	1500-row A1 generation gate.
docs/reports/phase1d_retriever_diversity_a1_mimo_v25_eval.md	Main Results; completed evaluations	1500-row A1 answer-label validation, 600-row stratified context labels, context reward candidate, and selector sweeps.
docs/reports/local_qwen_kv_estimates.md	KV-cache motivation; Appendix D	Analytical Qwen KV-cache memory table.

Reports that only contain raw command templates, intermediate validation summaries, or ignored local artifact paths are not copied verbatim. Their stable numbers are consolidated through the curated reports listed above.

References

[1] Yuntao Bai, Saurav Kadavath, Sandipan Kundu, Amanda Askell, Jackson Kernion, Andy Jones, Anna Chen, Anna Goldie, Azalia Mirhoseini, Cameron McKinnon, Carol Chen, Catherine Olsson, Chris Olah, Danny Hernandez, Dawn Drain, Deep Ganguli, Dustin Li, Eli Tran-Johnson, Ethan Perez, Jamie Kerr, Jared Mueller, Jeffrey Ladish, Joshua Landau, Kamal Ndousse, Kamile Lukosuite, Liane Lovitt, Michael Sellitto, Nelson Elhage, Nicholas Schiefer, Nova DasSarma Mercado, Robert Lasenby, Robin Larson, Sam Ringer, Scott Johnston, Shauna Kravec, Sheer Showk, Stanislav Fort, Tamera Lanham, Timothy Telleen-Lawton, Tom Conerly, Tom Henighan, Tristan Hume, Samuel R. Bowman, Zac Hatfield-Dodds, Ben Mann, Dario Amodei, Nicholas Joseph, Sam McCandlish, Tom Brown, and Jared Kaplan. Constitutional ai: Harmlessness from ai feedback. *arXiv preprint arXiv:2212.08073*, 2022. https://arxiv.org/abs/2212.08073.

[2] Zefan Cai, Yichi Zhang, Bofei Gao, Yuliang Liu, Tianyi Liu, Keming Lu, Wayne Xiong, Yue Dong, Baobao Chang, Junjie Hu, and Wen Xiao. Pyramidkv: Dynamic kv cache compression based on pyramidal information funneling. *arXiv preprint arXiv:2406.02069*, 2024. https://arxiv.org/abs/2406.02069.

[3] Gordon V. Cormack, Charles L. A. Clarke, and Stefan Buettcher. Reciprocal rank fusion outperforms condorcet and individual rank learning methods. In *Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 758–759, 2009. https://doi.org/10.1145/1571941.1572114.

[4] Darren Edge, Ha Trinh, Newman Cheng, Joshua Bradley, Alex Chao, Apurva Mody, Steven Truitt, and Jonathan Larson. From local to global: A graph rag approach to query-focused summarization. *arXiv preprint arXiv:2404.16130*, 2024. https://arxiv.org/abs/2404.16130.

[5] Shahul Es, Jithin James, Luis Espinosa-Anke, and Steven Schockaert. Ragas: Automated evaluation of retrieval augmented generation. *arXiv preprint arXiv:2309.15217*, 2023. https://arxiv.org/abs/2309.15217.

[6] Huiqiang Jiang, Qian Wu, Chin-Yew Lin, Yuqing Yang, and Lili Qiu. Llmlingua: Compressing prompts for accelerated inference of large language models. *arXiv preprint arXiv:2310.05736*, 2023. https://arxiv.org/abs/2310.05736.

[7] Huiqiang Jiang, Qian Wu, Xufang Luo, Dongsheng Li, Chin-Yew Lin, Yuqing Yang, and Lili Qiu. Longllmlingua: Accelerating and enhancing llms in long context scenarios via prompt compression. *arXiv preprint arXiv:2310.06839*, 2023. https://arxiv.org/abs/2310.06839.

[8] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Kuttler, Mike Lewis, Wen-tau Yih, Tim Rocktaschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive nlp tasks. *arXiv preprint arXiv:2005.11401*, 2020. https://arxiv.org/abs/2005.11401.

[9] Lihong Li, Wei Chu, John Langford, and Robert E. Schapire. A contextual-bandit approach to personalized news article recommendation. *arXiv preprint arXiv:1003.0146*, 2010. https://arxiv.org/abs/1003.0146.

[10] Yucheng Li, Bo Dong, Frank Guerin, and Chenghua Lin. Compressing context to enhance inference efficiency of large language models. *arXiv preprint arXiv:2310.06201*, 2023. https://arxiv.org/abs/2310.06201.

[11] Yuhong Li, Yingbing Huang, Bowen Yang, Bharat Venkitesh, Andrea Locatelli, Hanchen Ye, Tian Cai, Patrick Lewis, and Deming Chen. Snapkv: Llm knows what you are looking for before generation. *arXiv preprint arXiv:2404.14469*, 2024. https://arxiv.org/abs/2404.14469.

[12] Zirui Liu, Jiayi Yuan, Hongye Jin, Shaochen Zhong, Zhaozhuo Xu, Vladimir Braverman, Beidi Chen, and Xia Hu. Kivi: A tuning-free asymmetric 2bit quantization for kv cache. *arXiv preprint arXiv:2402.02750*, 2024. https://arxiv.org/abs/2402.02750.

[13] Rafael Rafailov, Archit Sharma, Eric Mitchell, Stefano Ermon, Christopher D. Manning, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model. *arXiv preprint arXiv:2305.18290*, 2023. https://arxiv.org/abs/2305.18290.

[14] Nandan Thakur, Nils Reimers, Andreas Ruckle, Abhishek Srivastava, and Iryna Gurevych. Beir: A heterogeneous benchmark for zero-shot evaluation of information retrieval models. *arXiv preprint arXiv:2104.08663*, 2021. https://arxiv.org/abs/2104.08663.

[15] Guangxuan Xiao, Yuandong Tian, Beidi Chen, Song Han, and Mike Lewis. Efficient streaming language models with attention sinks. *arXiv preprint arXiv:2309.17453*, 2023. https://arxiv.org/abs/2309.17453.

[16] Zhenyu Zhang, Ying Sheng, Tian Zhou, Tian Chen, Lianmin Zheng, Ruisi Cai, Zhao Song, Yuandong Tian, Christopher Re, Clark Barrett, Zhangyang Wang, and Beidi Chen. H2o: Heavy-hitter oracle for efficient generative inference of large language models. *arXiv preprint arXiv:2306.14048*, 2023. https://arxiv.org/abs/2306.14048.